Discrete Mathematics - Complete PYQ Answer Guide

UNIT I: SETS & PROPOSITIONS

1. Set Operations (Symmetric Difference, Union, Intersection, Subset, Power Set, Universal Set, Complement)

Union of Sets ($A \cup B$): Set of all elements belonging to A or B or both.

$A = \{1,2,3\}$, $B = \{3,4,5\} \rightarrow A \cup B = \{1,2,3,4,5\}$

Intersection of Sets ($A \cap B$): Set of all elements common to both A and B.

$A = \{1,2,3\}$, $B = \{3,4,5\} \rightarrow A \cap B = \{3\}$

Symmetric Difference ($A \triangle B$): Elements in A or B but not in both.

Formula: $A \triangle B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

$A = \{1,2,3\}$, $B = \{3,4,5\} \rightarrow A \triangle B = \{1,2,4,5\}$

Subset ($A \subseteq B$): Every element of A is also in B.

$A = \{1,2\}$, $B = \{1,2,3\} \rightarrow A \subseteq B$

Proper Subset ($A \subset B$): $A \subseteq B$ and $A \neq B$.

Universal Set (U): The set containing all elements under consideration.

If we discuss natural numbers, $U = \mathbb{N}$

Complement of A (A'): All elements in U but not in A.

$U = \{1,2,3,4,5\}$, $A = \{1,2\} \rightarrow A' = \{3,4,5\}$

Power Set P(A): Set of ALL subsets of A.

A = {1,2} → P(A) = {∅, {1}, {2}, {1,2}} → $|P(A)| = 2^n$ where n = |A|

UNIT II: RELATIONS & FUNCTIONS

(Covered inline with Graph/Algebraic structure questions below)

UNIT III: TREES

2. Binary Tree

A rooted tree where each node has at most 2 children (left and right).

```
      10
     /  \
    5    15
   / \
  3   7
```

Properties:

Max nodes at level i = 2^i

Max nodes in tree of height h = 2^(h+1) − 1

Min height for n nodes = $\lfloor \log_2 n \rfloor$

3. Binary Search Tree (BST)

A binary tree where:

Left subtree contains nodes with keys < root

Right subtree contains nodes with keys > root

Each subtree is also a BST

Example: Insert 50, 30, 70, 20, 40:

```
    50
   / \
  30   70
 / \
20  40
```

Inorder traversal of BST gives sorted output: 20, 30, 40, 50, 70

4. Ordered Tree

A rooted tree where the children of each node are ordered (left-to-right sequence matters).

Example: A family tree where eldest child always appears first — swapping children gives a different ordered tree even if same nodes exist.

5. Level and Height of a Tree

Level of a node: Number of edges from root to that node. Root is at level 0.

Height of a tree: Maximum level among all nodes (length of longest path from root to leaf).

```
    A      ← Level 0
   / \
  B   C    ← Level 1
 / \
D   E      ← Level 2
```

Height = 2

6. Regular m-ary Tree

A tree where every internal node has exactly m children.

Regular binary tree (m=2): Every internal node has exactly 2 children.

```
    A
   / \
  B   C
 / \ / \
D  E F  G
```

Formula: If T is a full m-ary tree with i internal nodes → total nodes $n = mi + 1$, leaves $l = (m-1)i + 1$

7. Eccentricity of a Vertex

Eccentricity e(v): Maximum distance from vertex v to any other vertex in the graph.

In a path graph A–B–C–D:

e(A) = 3, e(B) = 2, e(C) = 2, e(D) = 3

Radius = min eccentricity = 2, Diameter = max eccentricity = 3

8. Spanning Tree

A subgraph of a connected graph G that:

Is a tree (connected + acyclic)

Contains all vertices of G

Has exactly n−1 edges (n = number of vertices)

```
Graph:        Spanning Tree:
A–B–C          A–B–C
|  |   →     |
D–––+          D
```

A graph can have multiple spanning trees.

9. Cutset

A cutset is a minimal set of edges whose removal disconnects the graph (increases number of connected components).

"Minimal" means no proper subset of it is also a cutset.

Fundamental Cutset: A cutset associated with a spanning tree — it contains exactly one tree edge and possibly some non-tree edges.

10. Fundamental Circuit

When a non-tree edge is added to a spanning tree, it creates exactly one cycle — this cycle is called a fundamental circuit.

If spanning tree has edges {e1,e2,e3} and we add edge e4, the unique cycle formed = fundamental circuit of e4.

11. Forest

A forest is an acyclic graph (graph with no cycles). It is a collection of disjoint trees.

A tree is a connected forest. If a forest has n vertices and k trees → it has $n-k$ edges.

```
 A  D−E
/ \   \
B  C   F
```

(Tree 1) (Tree 2) → Together = Forest

12. Game Tree

A rooted tree used to represent all possible moves in a two-player game (like Tic-Tac-Toe, Chess).

Root = initial game state

Each level alternates between players (MAX and MIN)

Leaves = terminal states (win/loss/draw)

Used in Minimax algorithm

13. Prefix Code (Huffman)

A set of binary codes where no code is a prefix of another. Used in data compression.

Huffman Tree: Built by repeatedly merging two lowest-frequency nodes.

CharFreqCodeA50B210C1110D1111

No code is prefix of another → valid prefix code!

UNIT IV: GRAPH THEORY

14. Types of Graphs — Complete Definitions

Complete Graph (K_n): Every pair of vertices is connected by an edge.

K_n has $n(n-1)/2$ edges

K_4 has 6 edges — every vertex connects to all others

Regular Graph: Every vertex has the same degree (k-regular).

In K_3, every vertex has degree 2 → 2-regular graph

Bipartite Graph: Vertices can be divided into two disjoint sets V_1 and V_2 such that every edge goes between V_1 and V_2 (no edges within same set).

Example: Job assignment — set of workers and set of jobs

Workers: {A,B}   Jobs: {X,Y,Z}

A−X, A−Y, B−Y, B−Z → Bipartite!

Complete Bipartite Graph (Km,n): Every vertex in V_1 is connected to every vertex in V_2.

$K_{2,3}$ has $2 \times 3 = 6$ edges

Acyclic Graph: A graph with no cycles. Trees are acyclic connected graphs.

Complement of a Graph ($\bar{G}$): Has same vertices as G; edge exists in $\bar{G}$ iff it does NOT exist in G.

Weighted Graph: Each edge has an associated weight/cost (e.g., distance on a map).

Factor of a Graph: A spanning subgraph of G (uses all vertices, subset of edges). A 1-factor is a perfect matching.

15. Planar Graph

A graph that can be drawn on a plane without edges crossing.

Euler's Formula: $V - E + F = 2$ (V=vertices, E=edges, F=faces including outer face)

K_4 is planar. K_5 and $K_{3,3}$ are non-planar (Kuratowski's Theorem).

Example:

K_4: V=4, E=6, F=4 → $4-6+4 = 2$ ✓

16. Eulerian Graph, Euler Path & Euler Circuit

Euler Path: A path that visits every edge exactly once.

Exists iff graph has exactly 0 or 2 vertices of odd degree

If 2 odd-degree vertices: path starts at one, ends at other

Euler Circuit: A closed Euler path (starts and ends at same vertex).

Exists iff graph is connected and all vertices have even degree

Memory trick: Euler = Edges (every edge once)

Example:

A–B–C–A–D–B → Euler path in a suitable graph

17. Hamiltonian Path & Hamiltonian Circuit

Hamiltonian Path: A path that visits every vertex exactly once.

Hamiltonian Circuit: A closed Hamiltonian path (returns to start).

Memory trick: Hamilton = Haltestellen (vertices/stops — every vertex once)

No simple condition like Euler's exists. Sufficient conditions:

Dirac's Theorem: If every vertex has degree ≥ n/2 → Hamiltonian circuit exists

Ore's Theorem: If deg(u)+deg(v) ≥ n for every non-adjacent pair → Hamiltonian circuit exists

18. Graph Coloring

Assignment of colors to vertices such that no two adjacent vertices share the same color.

Chromatic number $\chi(G)$: Minimum colors needed.

Graph$\chi(G)$Tree2K_nnBipartite2Odd cycle3

Four Color Theorem: Any planar graph can be colored with at most 4 colors.

Application: Map coloring, register allocation, scheduling

19. Adjacency Matrix & Incidence Matrix

Adjacency Matrix (A): n×n matrix where A[i][j] = 1 if edge exists between v_i and v_j, else 0.

Graph: A–B, B–C, A–C (triangle)

  A B C

A [0 1 1]

B [1 0 1]

C [1 1 0]

Incidence Matrix (M): n×m matrix (n=vertices, m=edges). M[i][j] = 1 if vertex v_i is incident on edge e_j.

Edges: e1=AB, e2=BC, e3=AC

e1 e2 e3

A [1 0 1]

B [1 1 0]

C [0 1 1]

UNIT V: COUNTING & ALGEBRAIC STRUCTURES

20. Properties of Binary Operation

A binary operation * on set S assigns to each ordered pair (a,b) a unique element a*b ∈ S.

Properties:

Closure: a,b ∈ S → a*b ∈ S

Associativity: (ab)c = a(bc)

Commutativity: ab = ba

Identity (e): ae = ea = a

Inverse: $aa^{-1} = a^{-1}a = e$

21. Algebraic Structures — Complete Hierarchy

Groupoid ⊂ Semigroup ⊂ Monoid ⊂ Group ⊂ Abelian Group

StructureClosureAssociativeIdentityInverseCommutativeGroupoid✓✗✗✗✗Semigroup✓✓✗✗✗Monoid✓✓✓✗✗Group✓✓✓✓✗Abelian Group✓✓✓✓✓

Groupoid: Set with a binary operation. Example: (ℕ, −) — subtraction on naturals (not associative)

Semigroup: Associative groupoid. Example: (ℕ, +) without 0 — addition is associative but no identity

Monoid: Semigroup with identity. Example: (ℕ, +) with 0 — identity is 0; (Σ*, concatenation) — identity is empty string

Group: Monoid where every element has an inverse.

Example: ($\mathbb{Z}$, +) — identity=0, inverse of a is −a

Abelian (Commutative) Group: Group where ab = ba.

Example: ($\mathbb{Z}$, +), ($\mathbb{R}${0}, ×)

22. Cyclic Group

A group G is cyclic if there exists an element $g \in G$ (called generator) such that every element of G can be written as g^n for some integer n.

Notation: $G = \langle g \rangle$

Example: $\mathbb{Z}_4 = \{0,1,2,3\}$ under addition mod 4 is cyclic with generator 1:

$1^1=1$, $1^2=2$, $1^3=3$, $1^4=0$ → generates entire group

Example: ($\mathbb{Z}$, +) is cyclic with generator 1 or −1

23. Subgroup

A subset H of a group G is a subgroup if H itself forms a group under the same operation.

Conditions (Subgroup Test):

H is non-empty

$a,b \in H \rightarrow ab \in H$ (closure)

$a \in H \rightarrow a^{-1} \in H$ (inverse)

Example: ($\mathbb{Z}$, +) has subgroup ($2\mathbb{Z}$, +) = $\{\ldots,-4,-2,0,2,4,\ldots\}$ (all even integers)

24. Homomorphism of Groups

A function $f: G \to G'$ is a group homomorphism if:

*$f(a * b) = f(a) ' f(b)$ for all $a,b \in G$

Monomorphism: Injective homomorphism

Epimorphism: Surjective homomorphism

Isomorphism: Bijective homomorphism ($G \cong G'$)

Example: $f: (\mathbb{Z}, +) \to (\mathbb{Z}_2, +)$ defined by $f(n) = n \bmod 2$

$f(a+b) = (a+b) \bmod 2 = (a \bmod 2) + (b \bmod 2) = f(a) + f(b)$ ✓

Kernel of f: $\ker(f) = \{a \in G \mid f(a) = e'\}$ — always a normal subgroup

25. Normal Subgroup

A subgroup H of G is normal ($H \trianglelefteq G$) if:

$gHg^{-1} = H$ for all $g \in G$ (i.e., left coset = right coset)

Every subgroup of an Abelian group is normal.

Kernel of any homomorphism is a normal subgroup.

26. Ring, Ring with Unity, Integral Domain, Field

Ring $(R, +, \cdot)$: A set with two operations where:

$(R, +)$ is an Abelian group

$(R, \cdot)$ is a semigroup (associative)

Distributive law holds: a·(b+c) = a·b + a·c

Example: (ℤ, +, ×), (M_{2x2}, +, ×) (2×2 matrices)

Ring with Unity: A ring that has a multiplicative identity (1).

(ℤ, +, ×) is a ring with unity (1 is the identity)

Integral Domain: A commutative ring with unity that has no zero divisors.

Zero divisor: a ≠ 0 and b ≠ 0 but a·b = 0

Example: (ℤ, +, ×) — if a·b = 0 then a=0 or b=0 ✓

Non-example: $\mathbb{Z}_6$ — since 2·3 = 6 ≡ 0 (mod 6) but 2≠0, 3≠0

Field: A commutative ring with unity where every non-zero element has a multiplicative inverse.

Examples: (ℚ, +, ×), (ℝ, +, ×), ($\mathbb{Z}_p$, +, ×) for prime p

Hierarchy:

Ring → Ring with Unity → Integral Domain → Field

PropertyRingRing w/ UnityIntegral DomainFieldAbelian under +✓✓✓✓Multiplicative identity✗✓✓✓Commutative ×✗✗✓✓No zero divisors✗✗✓✓Mult. inverse for all nonzero✗✗✗✓

27. Identity and Inverse (Properties of Algebraic Structures)

Identity Element (e): An element e in set S such that:

a * e = e * a = a, for all a ∈ S

Additive identity: 0 (since a + 0 = a)

Multiplicative identity: 1 (since a × 1 = a)

Inverse of element a: An element a^{-1} such that:

$a * a^{-1} = a^{-1} * a = e$

Additive inverse of 3 is −3 (since 3 + (−3) = 0)

Multiplicative inverse of 2 is ½ (since 2 × ½ = 1)

28. Group Codes

A group code is a binary code where the set of all codewords forms a group under componentwise XOR (addition mod 2).

If 000, 011, 101, 110 are codewords:

011 XOR 101 = 110 ✓ (also a codeword → closed)

Used in error detection/correction

QUICK FORMULA SHEET

TopicFormula/Key FactPower Set size$|P(A)| = 2^n$Complete graph edgesn(n−1)/2Euler Circuit conditionAll vertices even degreeEuler Path conditionExactly 0 or 2 odd-degree verticesSpanning tree edgesn−1 (n = vertices)Euler's planar formulaV − E + F = 2BST inorderGives sorted sequenceHuffmanGreedy, min frequency merged firstChromatic number K_nnFull m-ary treen = mi + 1, l = (m−1)i + 1Group order (Lagrange)|H| divides |G|

Exam Tip: For every definition question, follow: Define → State properties → Give example. That structure guarantees full marks even if you forget minor details.